

A Trace Observation on the Rank-Two Projection Sparsification Question

Automatic Math Discovery Smoke Test

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Source and Extracted Question

The source is Grigory Ivanov, *No-dimensional results of combinatorial convexity. Dimension strikes back*, arXiv:2602.20035. In Section 6, after a deterministic Rudelson-type conjecture for positive semidefinite matrices, the paper asks whether a linear number of summands can suffice under additional structure; the displayed example asks whether $k = Cd$ is possible when each A_i is an orthogonal projection of rank 2.

conceptual step of passing from S_∞ to S_p , but it does seem a promising way. The upper bound would be very interesting to obtain a deterministic analogue of Rudelson's result.

Conjecture 6.1. *Let $A_1, \dots, A_n \in \mathcal{M}_d$ be positive semidefinite and suppose that*

$$\lambda_1 A_1 + \dots + \lambda_n A_n = \text{Id}_d,$$

where the coefficients λ_i are nonnegative and summing up to one. Then for every $\varepsilon \in (0, 1)$ there exists a deterministic algorithm that selects indices $i_1, \dots, i_k \in [n]$ with $k \leq C(\varepsilon) d \ln d$ and outputs positive coefficients $\beta_1, \dots, \beta_k$ such that

$$\left\| \beta_1 A_{i_1} + \dots + \beta_k A_{i_k} - \text{Id}_d \right\|_{S_\infty} \leq \varepsilon.$$

As a separate problem, we would like to understand whether a linear number of summands suffices under additional structure. For example, is $k = Cd$ possible when each A_i is an orthogonal projection of rank 2?

Claim

Under the normalization stated immediately before the rank-two projection question, namely

$$A_i \in \mathcal{M}_d, \quad A_i \geq 0, \quad \lambda_i \geq 0, \quad \sum_{i=1}^n \lambda_i = 1, \quad \sum_{i=1}^n \lambda_i A_i = I_d,$$

the case in which every A_i is an orthogonal projection of rank 2 is vacuous in dimensions $d \neq 2$. In dimension $d = 2$, the problem is solved exactly with one summand.

Theorem 1. *Let $A_1, \dots, A_n \in \mathcal{M}_d$ be orthogonal projections of rank 2. If there are nonnegative scalars $\lambda_1, \dots, \lambda_n$ with $\sum_i \lambda_i = 1$ and*

$$\sum_{i=1}^n \lambda_i A_i = I_d,$$

then $d = 2$. Consequently, for the rank-two projection formulation as printed, there are no feasible instances in dimensions $d > 2$, and every feasible instance in dimension 2 is represented exactly by a single summand.

Proof. The trace of an orthogonal projection is its rank. Hence $\text{tr}(A_i) = 2$ for every i . Taking traces in the assumed identity gives

$$d = \text{tr}(I_d) = \text{tr} \left(\sum_{i=1}^n \lambda_i A_i \right) = \sum_{i=1}^n \lambda_i \text{tr}(A_i) = 2 \sum_{i=1}^n \lambda_i = 2.$$

Thus $d = 2$ is forced.

If $d = 2$, an orthogonal projection of rank 2 on the whole space is I_2 . Pick any index i_1 and set $\beta_1 = 1$. Then

$$\|\beta_1 A_{i_1} - I_2\|_{S_\infty} = 0.$$

Thus $k = 1$ suffices in the only feasible dimension. □

Interpretation and Limitation

This should be viewed as a normalization issue in the printed rank-two example, not as a solution to a renormalized sparsification problem. A different question, for example allowing the original convex coefficients to have total $d/2$, or replacing projections by positive semidefinite matrices normalized to have trace d , is not addressed by this packet.

Verification Notes

No computation is needed. The entire argument is the trace identity above. The main point for human review is to confirm that “orthogonal projection of rank 2” is meant in the standard sense, so that its trace equals 2.